

Section 12.5

1. 15 points Let $f(x) = 8x^3 - 12x^2 + 6x - 1$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[0, 1]$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

First, we find $f'(x)$:

$$f'(x) = 24x^2 - 24x + 6.$$

Then, we set $f'(x) = 0$ and solve to find the root $x = \frac{1}{2}$. Following this, we find $f''(x)$:

$$f''(x) = 48x - 24.$$

As such, $f''(\frac{1}{2}) = 0$, and f has an inflection point at $x = \frac{1}{2}$. Checking the endpoints, we then conclude that $(0, f(0)) = (0, -1)$ is indeed an absolute minimum of f on the interval $[0, 1]$, and $(1, f(1)) = (1, 1)$ is indeed an absolute maximum of f on the interval $[0, 1]$.

2. 15 points Let $f(x) = x^2 - 5$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[-2, 2]$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

First, we find $f'(x)$:

$$f'(x) = 2x.$$

Then, we set $f'(x) = 0$ and solve to find the root $x = 0$. Following this, we find $f''(x)$:

$$f''(x) = 2.$$

As such, $f''(0) = 2 > 0$, and f is concave upward at $x = 0$. Noting that 0 is indeed in $[-2, 2]$, we then conclude that $(0, f(0)) = (0, -5)$ is indeed an absolute minimum of f on the interval $[-2, 2]$. Checking the endpoints, we then conclude that $(-2, f(-2)) = (-2, -1)$ and $(2, f(2)) = (2, -1)$ are absolute maximums of f on the interval $[-2, 2]$.

3. 15 points Let $f(x) = x^2 - 5$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[1, 5]$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

First, we find $f'(x)$:

$$f'(x) = 2x.$$

Then, we set $f'(x) = 0$ and solve to find the root $x = 0$. Following this, we find $f''(x)$:

$$f''(x) = 2.$$

As such, $f''(0) = 2 > 0$, and f is concave upward at $x = 0$. Noting that 0 is not in $[1, 5]$, we disregard this point.

Checking the endpoints, we then reach that $(1, f(1)) = (1, -4)$ is an absolute minimum of f on the interval $[1, 5]$, and $(5, f(5)) = (5, 20)$ is an absolute maximum of f on the interval $[1, 5]$.

4. 15 points Let $f(x) = x^2 - 4x + 8$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[0, \infty)$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

First, we find $f'(x)$:

$$f'(x) = 2x - 4.$$

Then, we set $f'(x) = 0$ and solve to find the root $x = 2$. Following this, we find $f''(x)$:

$$f''(x) = 2.$$

As such, $f''(2) = 2 > 0$, and f is concave upward at $x = 2$. Noting that 2 is in $[0, \infty)$, we may indeed conclude that $(2, f(2)) = (2, 4)$ is an absolute minimum of f on $[0, \infty)$.

Checking the endpoint $x = 0$, we then reach $(0, f(0)) = (0, 8)$. However, this is not an absolute maximum, as $f(x)$ is symmetric about $x = 2$, and increases without bound as x approaches ∞ . As such, f has no absolute maximum on the interval $[0, \infty)$.

5. 15 points Let $f(x) = x - \frac{2}{x} + \frac{5}{x^3}$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $(0, \infty)$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

First, we find $f'(x)$:

$$f'(x) = 1 + \frac{2}{x^2} - \frac{15}{x^4}.$$

Then, we set $f'(x) = 0$ and solve to find the roots $x = \sqrt{3}$ and $x = -\sqrt{3}$. Following this, we find $f''(x)$:

$$f''(x) = -\frac{4}{x^3} + \frac{60}{x^5}.$$

As such, $f''(\sqrt{3}) = \frac{16}{3\sqrt{3}} > 0$, $f''(-\sqrt{3}) = -\frac{16}{3\sqrt{3}} < 0$, and we reach that f is concave upward at $x = \sqrt{3}$ and concave downward at $x = -\sqrt{3}$. Noting that $\sqrt{3}$ is indeed in $(0, \infty)$, we then conclude that $(\sqrt{3}, f(\sqrt{3})) = (\sqrt{3}, \frac{8\sqrt{3}}{9})$ is indeed an absolute minimum of f on the interval $(0, \infty)$.

As $-\sqrt{3}$ is not in the interval $(0, \infty)$, we disregard this point. As $(0, \infty)$ has no endpoints, we conclude that f has no absolute maximum on the interval $(0, \infty)$.

6. 15 points (bonus) Let $f(x) = \frac{x^6}{e^x}$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $(0, \infty)$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

First, we find $f'(x)$:

$$f'(x) = \frac{6x^5 e^x - x^6 e^x}{(e^x)^2} = \frac{x^5(6 - x)}{e^x}.$$

Then, we set $f'(x) = 0$ and solve to find the roots $x = 0$ and $x = 6$. Following this, we find $f''(x)$:

$$f''(x) = \frac{x^4(x^2 - 12x + 30)}{e^x}.$$

As such, $f''(0) = 0$, $f''(6) = \frac{6^4(6^2 - 12(6) + 30)}{e^6} = \frac{-7776}{e^6} < 0$, and we reach that f has an inflection point at $x = 0$ and is concave downward at $x = 6$. Noting that 6 is indeed in $(0, \infty)$, we then conclude that $(6, f(6)) = (6, \frac{46,656}{e^6})$ is indeed an absolute maximum of f on the interval $(0, \infty)$.

As $(0, \infty)$ has no endpoints, we conclude that f has no absolute minimum on the interval $(0, \infty)$.

Section 12.6

1. 20 points A company manufactures and sells x cellphones per week. The weekly price-demand and cost equations are given below.

$$p = 600 - 0.5x \text{ and } C(x) = 15,000 + 140x.$$

What price should the company charge for the phones, and how many phones should be produced, to maximize the weekly revenue? What is the maximum weekly revenue?

To find the revenue function $R(p)$, we first isolate x in the price-demand equation:

$$x = 1200 - 2p.$$

Then, we find the revenue function $R(p)$:

$$R(p) = 1200p - 2p^2$$

and the derivative of the revenue function, $R'(p)$:

$$R'(p) = 1200 - 4p.$$

Setting $R'(p) = 0$, we solve to find the root $p = 300$.

Now, we find $R''(p)$:

$$f''(x) = -4.$$

As such, $R''(300) = -4 < 0$, and we may conclude $(300, R(300))$ is an absolute maximum of $R(p)$ on the interval $(-\infty, \infty)$.

Thus, the company should price the cell phones at \$300 and manufacture $x = 1200 - 2(300) = 600$ cell phones.

2. 20 points A box is to be made at a manufacturing company out of a rectangular piece of cardboard 15 inches wide and 20 inches long. Squares of equal size will be cut out of each corner, and then the ends and sides will be folded up to form a rectangular box with an open top. What size square should be cut from each corner to maximize volume?

First, we set up our volume equation $V(x)$:

$$V(x) = (15 - 2x)(20 - 2x)x, \quad 0 \leq x \leq \frac{15}{2}.$$

Then, we find $V'(x)$:

$$V'(x) = 12x^2 - 140x + 300.$$

Setting $V'(x) = 0$, we solve to find the roots $x = \frac{35 - 5\sqrt{13}}{6}$ and $x = \frac{35 + 5\sqrt{13}}{6}$.

Now, we find $V''(x)$:

$$V''(x) = 24x - 140.$$

As such, $V''(\frac{35 - 5\sqrt{13}}{6}) = -20\sqrt{13} < 0$ and $V''(\frac{35 + 5\sqrt{13}}{6}) = 20\sqrt{13} > 0$, and we may conclude that V is concave downward at $x = \frac{35 - 5\sqrt{13}}{6}$ and concave upward at $\frac{35 + 5\sqrt{13}}{6}$. Noting that $\frac{35 - 5\sqrt{13}}{6}$ is in the interval $[0, \frac{15}{2}]$, we may conclude that $(\frac{35 - 5\sqrt{13}}{6}, V(\frac{35 - 5\sqrt{13}}{6}))$ is an absolute maximum of $V(x)$ on the interval $[0, \frac{15}{2}]$. Noting that $\frac{35 + 5\sqrt{13}}{6}$ is not in the interval $[0, \frac{15}{2}]$, we may disregard this point. Checking the endpoints, we find that $V(0) = 0$ and $V(\frac{15}{2}) = 0$.

As such, squares with edges of length $x = \frac{35 - 5\sqrt{13}}{6} \approx 2.829$ inches should be cut from the cardboard rectangle to maximize the volume of the boxes produced.

3. 20 points A deli sells 720 sandwiches per day at a price of \$6 each. A market survey shows that for every \$0.10 reduction in the price, 40 more sandwiches will be sold. How much should the deli charge in order to maximize revenue?

First, we set up our revenue equation $R(x)$:

$$R(x) = (720 + 40x)(6 - 0.1x).$$

Then, we find $R'(x)$:

$$R'(x) = 168 - 8x.$$

Setting $R'(x) = 0$, we solve to find the root $x = 21$.

Now, we find $R''(x)$:

$$R''(x) = -8.$$

As such $R''(21) = -8 < 0$, and we may conclude that $(21, R(21)) = (21, 6,084)$ is an absolute maximum of $R(x)$ on the interval $(-\infty, \infty)$.

As such, the deli should charge $\$(6 - 0.1(21)) = \3.90 per sandwich to maximize revenue.

Section 13.1

1. 10 points Find $\int \frac{6}{x} dx$

$$\begin{aligned}\int \frac{6}{x} dx &= 6 \int \frac{1}{x} dx \\ &= 6\ln(|x|) + c\end{aligned}$$

2. 10 points Find f such that $f'(x) = \frac{5}{\sqrt{x}}$ and $f(16) = 55$.

First, we find $\int f'(x) dx$:

$$\begin{aligned}\int \frac{5}{\sqrt{x}} dx &= 5 \int \frac{1}{\sqrt{x}} dx \\ &= 5 \int x^{-\frac{1}{2}} dx \\ &= 5(2x^{\frac{1}{2}}) + c \\ &= 10x^{\frac{1}{2}} + c.\end{aligned}$$

Then, we solve for c :

$$\begin{aligned}10(16)^{\frac{1}{2}} + c &= 55 \\ 10(4) + c &= 55 \\ c &= 15.\end{aligned}$$

Thus, $f(x) = 10x^{\frac{1}{2}} + 15$.

3. 10 points Find y such that $\frac{dy}{dt} = 5e^t - 4$ and $y(0) = 4$.

First, we find $\int \frac{dy}{dt} dt$:

$$\begin{aligned}\int 5e^t - 4 dt &= 5 \int e^t dt - \int 4 dt \\ &= 5e^t - 4t + c.\end{aligned}$$

Then, we solve for c :

$$\begin{aligned}5e^0 - 4(0) + c &= 4 \\ 5 + c &= 4 \\ c &= -1.\end{aligned}$$

Thus, $y(t) = 5e^t - 4t - 1$.

4. 10 points If the marginal profit of producing x surfboards is given by $P'(x) = 40x - x^2$, and $P(10) = 20$ where $P(x)$ is the profit in dollars, find $P(x)$.

First, we find $\int P'(x) dx$:

$$\int 40x - x^2 dx = 20x^2 - \frac{1}{3}x^3 + c.$$

Then, we solve for c :

$$\begin{aligned}20(10)^2 - \frac{1}{3}(10)^3 + c &= 20 \\ 2,000 - \frac{1,000}{3} + c &= 20 \\ c &= 20 - 2,000 + \frac{1,000}{3} \\ c &= -\frac{4,940}{3}.\end{aligned}$$

Thus, $P(x) = 20x^2 - \frac{1}{3}x^3 - \frac{4,940}{3}$.

5. 10 points Find $\int \frac{x^3 e^x - 3x^2}{x^3} dx$.

$$\begin{aligned}\int \frac{x^3 e^x - 3x^2}{x^3} dx &= \int e^x - 3x^{-1} dx \\ &= e^x - 3\ln(|x|) + c\end{aligned}$$

6. 15 points (bonus) Find $\int \frac{d}{dx}(x^4 + 5x^2 + 8) dx$.

$$\begin{aligned}\int \frac{d}{dx}(x^4 + 5x^2 + 8) dx &= \int 4x^3 + 10x dx \\ &= x^4 + 5x^2 + c\end{aligned}$$

Section 13.2

1. In each problem, find $\int f(x) dx$.

(a) 10 points $f(x) = 2x(x^2 + 2)^{11}$

Let $u = x^2 + 2$. Then, $du = 2x dx$. Substituting, we yield

$$\begin{aligned}\int 2x(x^2 + 2)^{11} dx &= \int u^{11} du \\ &= \frac{1}{12} u^{12} + c \\ &= \frac{1}{12} (x^2 + 2)^{12} + c.\end{aligned}$$

(b) 10 points $f(x) = 10e^{10x}$

Let $u = 10x$. Then, $du = 10 dx$. Substituting, we yield

$$\begin{aligned}\int 10e^{10x} dx &= \int e^u du \\ &= e^u + c \\ &= e^{10x} + c.\end{aligned}$$

(c) 10 points $f(x) = x^{12}e^{x^{13}}$

Let $u = x^{13}$. Then, $du = 13x^{12} dx$ and $\frac{1}{13} du = x^{12} dx$. Substituting, we yield

$$\begin{aligned}\int x^{12}e^{x^{13}} dx &= \int \frac{1}{13} e^u du \\ &= \frac{1}{13} e^u + c \\ &= \frac{1}{13} e^{x^{13}} + c.\end{aligned}$$

(d) 10 points $f(x) = \frac{x^4}{(6 - x^5)^2}$

Let $u = 6 - x^5$. Then, $du = -5x^4 dx$ and $-\frac{1}{5}du = x^4 dx$. Substituting, we yield

$$\begin{aligned}\int \frac{x^4}{(6 - x^5)^2} dx &= \int -\frac{1}{5} \frac{1}{u^2} du \\ &= \left(-\frac{1}{5}\right) \left(-\frac{1}{u}\right) + c \\ &= \frac{1}{5u} + c.\end{aligned}$$

2. 10 points Find $\int x\sqrt{12 - x^2} dx$.

Let $u = 12 - x^2$. Then, $du = -2x dx$ and $-\frac{1}{2}du = x dx$. Substituting, we yield

$$\begin{aligned}\int x\sqrt{12 - x^2} dx &= \int -\frac{1}{2}\sqrt{u} du \\ &= -\frac{1}{2} \left(\frac{2}{3} u^{\frac{3}{2}}\right) + c \\ &= -\frac{2}{6} u^{\frac{3}{2}} + c\end{aligned}$$

3. 10 points If the marginal cost for producing x of a certain item is given by $C'(x) = 8(x+10)^2$, and the fixed cost is $C(0) = 2,000$ (in dollars), find the cost function $C(x)$.

First, we solve $\int C'(x)dx$:

Let $u = x + 10$. Then, $du = dx$. Substituting, we yield

$$\begin{aligned}\int 8(x+10)^2 dx &= \int 8u^2 du \\ &= \frac{8}{3}u^3 + c \\ &= \frac{8}{3}(x+10)^3 + c.\end{aligned}$$

Now, we solve for c :

$$\begin{aligned}\frac{8}{3}((0)+10)^3 + c &= 2,000 \\ \frac{8,000}{3} + c &= 2,000 \\ c &= -\frac{2,000}{3}.\end{aligned}$$

Thus, $C(x) = \frac{8}{3}(x+10)^3 - \frac{2,000}{3}$.

Section 13.4

1. 10 points Calculate $\int_4^9 9x + x^2 dx$, given that

$$\int_4^9 x^2 dx = \frac{665}{3}, \text{ and } \int_4^9 x dx = \frac{65}{2}.$$

$$\begin{aligned}\int_4^9 9x + x^2 dx &= 9 \int_4^9 x dx + \int_4^9 x^2 dx \\ &= 9 \frac{65}{2} + \frac{665}{3} \\ &= \frac{3085}{6}\end{aligned}$$

2. 10 points Calculate $\int_1^{10} 5x^2 + x dx$ given that

$$\int_1^9 x^2 dx = \frac{728}{3}, \int_9^{10} x^2 dx = \frac{271}{3}, \text{ and } \int_1^{10} x dx = \frac{99}{2}.$$

$$\begin{aligned}\int_1^{10} 5x^2 + x dx &= 5 \int_1^{10} x^2 dx + \int_1^{10} x dx \\ &= 5 \left(\int_1^9 x^2 dx + \int_9^{10} x^2 dx \right) + \int_1^{10} x dx \\ &= 5 \left(\frac{728}{3} + \frac{271}{3} \right) + \frac{99}{2} \\ &= \frac{3,429}{2}\end{aligned}$$

3. 10 points Let $f(x) = x^2 + 1$. Over the interval $[0, 4]$, calculate R_4 , the right-hand Riemann sum.

First, we note that the interval $[0, 4]$ has length 4. Then, we divide the length of the interval by the number of rectangles in our approximation to yield $\Delta x = \frac{4}{4} = 1$. Then, we calculate R_4 :

$$\begin{aligned} R_4 &= \Delta x \left(\sum_{k=1}^n f(0 + k\Delta x) \right) \\ &= (1)(f(0 + 1) + f(0 + 2) + f(0 + 3) + f(0 + 4)) \\ &= f(1) + f(2) + f(3) + f(4) \\ &= ((1)^2 + 1) + ((2)^2 + 1) + ((3)^2 + 1) + ((4)^2 + 1) \\ &= 2 + 5 + 10 + 17 \\ &= 34. \end{aligned}$$

4. 10 points Let $f(x) = x^2 + 1$. Over the interval $[0, 4]$, calculate L_4 , the left-hand Riemann sum.

First, we note that the interval $[0, 4]$ has length 4. Then, we divide the length of the interval by the number of rectangles in our approximation to yield $\Delta x = \frac{4}{4} = 1$. Then, we calculate R_4 :

$$\begin{aligned} L_4 &= \Delta x \left(\sum_{k=1}^n f(4 - k\Delta x) \right) \\ &= (1)(f(4 - 1) + f(4 - 2) + f(4 - 3) + f(4 - 4)) \\ &= f(3) + f(2) + f(1) + f(0) \\ &= ((3)^2 + 1) + ((2)^2 + 1) + ((1)^2 + 1) + ((0)^2 + 1) \\ &= 10 + 5 + 2 + 1 \\ &= 18. \end{aligned}$$

Section 13.5

1. 10 points Calculate $\int_2^3 6x^2(x^3 + 2)^{-2} dx$.

First, we calculate $\int 6x^2(x^3 + 2)^{-2} dx$:

Let $u = x^3 + 2$. Then, $du = 3x^2 dx$ and $2du = 6x^2 dx$. Substituting, we yield

$$\begin{aligned}\int 6x^2(x^3 + 2)^{-2} dx &= \int 2u^{-2} du \\ &= -2u^{-1} + c \\ &= -2(x^3 + 2)^{-1} + c.\end{aligned}$$

Then, we find $\int_2^3 6x^2(x^3 + 2)^{-2} dx$:

$$\begin{aligned}\int_2^3 6x^2(x^3 + 2)^{-2} dx &= (-2((3)^3 + 2)^{-1} + c) - (-2((2)^3 + 2)^{-1} + c) \\ &= -\frac{2}{29} + c + \frac{2}{10} - c \\ &= -\frac{2}{29} + \frac{2}{10} = \frac{19}{145}.\end{aligned}$$

2. 10 points Calculate $\int_1^{10} 3x^2 + 2x + 1 dx$.

First, we calculate $\int 3x^2 + 2x + 1 dx$:

$$\int 3x^2 + 2x + 1 dx = x^3 + x^2 + x + c.$$

Then, we find $\int_2^3 6x^2(x^3 + 2)^{-2} dx$:

$$\begin{aligned}\int_1^{10} 3x^2 + 2x + 1 dx &= (10^3 + 10^2 + 10 + c) - (1^3 + 1^2 + 1 + c) \\ &= 1110 + c - 3 - c = 1107.\end{aligned}$$

3. 10 points Calculate $\int_0^3 2e^{2x} \, dx$.

First, we calculate $\int 2e^{2x} \, dx$:

$$\int 2e^{2x} \, dx = e^{2x} + c.$$

Then, we find $\int_0^3 2e^{2x} \, dx$:

$$\begin{aligned} \int_0^3 2e^{2x} \, dx &= \left(e^{2(3)} + c \right) - \left(e^{2(0)} + c \right) \\ &= e^6 + c - 1 - c \\ &= e^6 - 1. \end{aligned}$$

4. 10 points Calculate $\int_{-2}^2 x^3 \, dx$.

First, we calculate $\int x^3 \, dx$:

$$\int x^3 \, dx = \frac{1}{4}x^4 + c.$$

Then, we find $\int_{-2}^2 x^3 \, dx$:

$$\begin{aligned} \int_{-2}^2 x^3 \, dx &= \left(\frac{1}{4}(2)^4 + c \right) - \left(\frac{1}{4}(-2)^4 + c \right) \\ &= \frac{64}{2}. \end{aligned}$$

5. 10 points Calculate the average value of $f(x) = 5 - x^2$ over the interval $[-3, 2]$.

First, we calculate $\int 5 - x^2 \, dx$:

$$\int 5 - x^2 \, dx = 5x - \frac{1}{3}x^3 + c.$$

Then, we find $\int_{-3}^2 5 - x^2 \, dx$:

$$\begin{aligned} \int_{-3}^2 5 - x^2 \, dx &= \left(5(2) - \frac{1}{3}(2)^3 + c \right) - \left(5(-3) - \frac{1}{3}(-3)^3 + c \right) \\ &= \frac{40}{3}. \end{aligned}$$

Then, we calculate $\frac{1}{2 - (-3)} \int_{-3}^2 5 - x^2 \, dx$:

$$\frac{1}{2 - (-3)} \int_{-3}^2 5 - x^2 \, dx = \left(\frac{1}{5} \right) \left(\frac{40}{3} \right) = \frac{40}{15}.$$

6. 10 points Calculate the average value of $f(x) = 32e^{0.04x}$ over the interval $[10, 30]$.

First, we calculate $\int 32e^{0.04x} dx$:

Let $u = 0.04x$. Then, $du = 0.04dx$ and $800du = 32dx$. Substituting, we yield

$$\begin{aligned}\int 32e^{0.04x} dx &= \int 800e^u du \\ &= 800e^u + c \\ &= 800e^{0.04x} + c.\end{aligned}$$

Then, we find $\int_{10}^{30} 32e^{0.04x} dx$:

$$\begin{aligned}\int_{10}^{30} 32e^{0.04x} dx &= \left(800e^{0.04(30)} + c\right) - \left(800e^{0.04(10)} + c\right) \\ &= 800e^{\frac{6}{5}} + c - 800e^{\frac{2}{5}} - c \\ &= 800 \left(e^{\frac{6}{5}} - e^{\frac{2}{5}}\right).\end{aligned}$$

Then, we calculate $\frac{1}{30 - (10)} \int_{10}^{30} 32e^{0.04x} dx$:

$$\frac{1}{20} \int_{10}^{30} 32e^{0.04x} dx = \left(\frac{1}{20}\right) \left(800 \left(e^{\frac{6}{5}} - e^{\frac{2}{5}}\right)\right) = 40 \left(e^{\frac{6}{5}} - e^{\frac{2}{5}}\right).$$